

A Node-indexed Construction of the Extended Bearing Rigidity Matrix for Heterogeneous Networks

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Abstract

The unified framework of [1] tests infinitesimal bearing rigidity by comparing the rank of an extended rigidity matrix with the rank of the same matrix built on the complete graph. While applying it to formations that mix ground vehicles with aerial platforms, we obtained rank comparisons that disagreed with rigidity, and this note reports what we believe is the cause. In eq. (10) of [1] the factor $\mathbf{U}$ is indexed by edges and multiplies the incidence matrix from the left, so within a row block it acts on the difference of the two endpoint variations and imposes the same restriction on both of them. When the endpoints lie in different manifolds, particularly in one planar and one spatial, it is not possible to find a choice of $\mathbf{U}_{ij}$ that both reproduces the bearing differential and vanishes on the inadmissible directions of the two endpoints. The two requirements are compatible only when the agents are vertically aligned. Definition 13 of [1] is still satisfied, since it constrains the matrix only on admissible variations, but the rank then carries a contribution that depends on the edge set, and it is ranks that Theorem 2 compares. We give a minimal example checkable by hand, a three-agent formation, that the comparison reports as flexible although it is rigid. Restricting the degrees of freedom node-wise rather than edge-wise removes the difficulty, and the resulting matrix coincides with eq. (10) of [1] for every homogeneous formation, so Table I and Section V of [1] are unaffected. The rigidity conclusions of the two case studies there are also unaffected, since both use the complete graph. A node-wise restriction of this kind is already used in the heterogeneous formation control formulation of [2], which supports the reading proposed here. All statements are checked against a finite-difference Jacobian of the bearing map.

1 Summary

Consider three agents at

$$p_1 = (0, 0, 0), \quad p_2 = (2, 0, 0), \quad p_3 = (1, 1, 1), \quad (1)$$

with agents 1 and 2 in $\mathcal{D}_1 = \mathcal{D}_2 = \mathbb{R}^2$ and agent 3 in $\mathcal{D}_3 = \mathbb{R}^3$, and with $R_i = I_3$, the attitudes being constant since neither domain has rotational freedom. Let $\mathcal{G}$ consist of the two edges (v_1, v_3) and (v_2, v_3) . The formation has $c = 7$ degrees of freedom, and Section 4 shows that its bearing-preserving motions are exactly two horizontal translations and uniform scaling, for $\mathcal{G}$ and for the complete graph $\mathcal{K}$. It is therefore infinitesimally bearing rigid.

	rk $\mathbf{B}_{\mathcal{G}}$	rk $\mathbf{B}_{\mathcal{K}}$	Theorem 2 concludes
$\mathbf{B}_{\mathcal{G}}^{\times}$, eq. (10) of [1]	4	5	flexible
$\mathbf{B}_{\mathcal{G}}^{\bullet}$, eq. (9) below	4	4	rigid
finite differences of $b_{\mathcal{G}}$	4	4	rigid

The extra unit of rank in case of $\mathbf{B}_{\mathcal{G}}^{\times}$ on the complete graph is associated with a vertical motion of the two planar agents. A single factor $\mathbf{U}_{ij}$ sits between $\mathbf{D}_p$ and $\bar{\mathbf{E}}^{\top}$, so it acts on the

difference of the two endpoint variations and applies the same restriction to both endpoints, while a planar and a spatial agent need different ones. Section 3 writes the products out block by block.

The matrix $\mathbf{B}_G^\times$ reproduces the bearing differential on admissible variations, so Definition 13 of [1] is met, and the ranks of the two constructions agree once both are restricted to motions the agents can perform. The two are identical for every homogeneous formation, and more generally whenever the two endpoints of each edge have the same translational projector (Proposition 2), so Table I and Section V of [1] are untouched. Both case studies of [1] use complete graphs, where $\text{rk } \mathbf{B}_G = \text{rk } \mathbf{B}_K$ holds trivially. The paper also studies proper subgraphs $\mathcal{G}_1$ and $\mathcal{G}_2$ in the heterogeneous case, but its conclusions about those subgraphs follow from explicit infinitesimal variations rather than from a rank comparison. Seeing a discrepancy requires a heterogeneous formation on an incomplete graph (that includes an edge between a planar and spatial agent).

2 Notation

We follow [1]. Agent i has position p_i , attitude R_i and $c_i = \dim \mathcal{D}_i$ degrees of freedom, and $c = \sum_i c_i$. An edge $e_k = (v_i, v_j)$ carries the bearing measured by i of j ,

$$b_{ij} = R_i^\top \hat{p}_{ij}, \quad \hat{p}_{ij} = \frac{p_j - p_i}{\|p_j - p_i\|}, \quad s_{ij} = \|p_j - p_i\|^{-1}. \quad (2)$$

We differentiate (2) in the two factors separately, since the two carry the two halves of the rigidity matrix. For the direction, with $p_{ij} = p_j - p_i$,

$$\begin{aligned} \frac{d}{dt} \hat{p}_{ij} &= \frac{\dot{p}_{ij}}{\|p_{ij}\|} - \frac{p_{ij} (p_{ij}^\top \dot{p}_{ij})}{\|p_{ij}\|^3} = s_{ij} (I_3 - \hat{p}_{ij} \hat{p}_{ij}^\top) \dot{p}_{ij} \\ &= s_{ij} P(\hat{p}_{ij}) (\dot{p}_j - \dot{p}_i), \quad P(x) = I_3 - xx^\top. \end{aligned} \quad (3)$$

For the attitude, with ω_i the angular velocity in the world frame we have $\dot{R}_i = [\omega_i]_\times R_i$, hence $\dot{R}_i^\top = R_i^\top [\omega_i]_\times^\top = -R_i^\top [\omega_i]_\times$ and

$$\dot{R}_i^\top \hat{p}_{ij} = -R_i^\top [\omega_i]_\times \hat{p}_{ij} = -R_i^\top (\omega_i \times \hat{p}_{ij}) = +R_i^\top (\hat{p}_{ij} \times \omega_i) = R_i^\top [\hat{p}_{ij}]_\times \omega_i. \quad (4)$$

Adding (3) and (4),

$$\dot{b}_{ij} = \underbrace{s_{ij} R_i^\top P(\hat{p}_{ij}) (\dot{p}_j - \dot{p}_i)}_{\mathbf{D}_p^{(k)}} - \underbrace{(-R_i^\top [\hat{p}_{ij}]_\times)}_{\mathbf{D}_a^{(k)}} \omega_i, \quad (5)$$

which is eq. (1) of [1] with the sign of $\mathbf{D}_a^{(k)}$ as defined there. The minus in front of $\mathbf{D}_a^{(k)}$ is the one supplied by $\bar{\mathbf{E}}_o^\top$ in Section 3.

Let S_i and P_i be the orthogonal projectors onto the translational and rotational directions agent i can realise, so that $\text{rk } S_i + \text{rk } P_i = c_i$.

$\mathcal{D}_i$	$\mathbb{R}^2$	$\mathbb{R}^3$	$\mathbb{R}^2 \times S^1$	$\mathbb{R}^3 \times S^1$	$\text{SE}(3)$
S_i	$\text{diag}(1, 1, 0)$	I_3	$\text{diag}(1, 1, 0)$	I_3	I_3
P_i	0	0	$e_3 e_3^\top$	$vv^\top$	I_3

A variation δ is lifted to $\delta^+ \in \mathbb{R}^{6n}$ by zero padding, as in (8) of [1]. The lifted variations fill the *admissible subspace*

$$\mathcal{A} = \text{range } \bar{S} \oplus \text{range } \bar{P} \subseteq \mathbb{R}^{6n}, \quad \bar{S} = \text{blkdiag}(S_i), \quad \bar{P} = \text{blkdiag}(P_i), \quad (6)$$

of dimension c . Let $A \in \mathbb{R}^{6n \times c}$ have orthonormal columns spanning $\mathcal{A}$, and take the variation coordinates to be the ones it induces, so that

$$\delta^+ = A\delta. \quad (7)$$

We write $q_v := 6n - c$ for the dimension of $\mathcal{A}^\perp$. Finally let $J_{\mathcal{G}} \in \mathbb{R}^{3m \times c}$ denote the Jacobian of the bearing map in these coordinates, so that $J_{\mathcal{G}}\delta = \dot{b}_{\mathcal{G}}$ whenever $\delta^+ = A\delta$, with $\dot{b}_{\mathcal{G}}$ stacking the row blocks (5).

3 Where the DOF restriction enters the product

We write $\mathbf{B}_{\mathcal{G}}^\times$ for the extended matrix of Proposition 2 of [1] and $\mathbf{B}_{\mathcal{G}}^\bullet$ for the node-indexed construction, and use these names throughout. The first is

$$\mathbf{B}_{\mathcal{G}}^\times = [\mathbf{D}_p \mathbf{U} \bar{\mathbf{E}}^\top \quad \mathbf{D}_a \mathbf{V} \bar{\mathbf{E}}_o^\top], \quad \mathbf{U} = \text{diag}(\mathbf{U}_{ij}), \quad \mathbf{V} = \text{diag}(\mathbf{V}_{ij}), \quad (8)$$

and the second is the same object with the two factors moved to the right of the incidence matrices and indexed by node,

$$\mathbf{B}_{\mathcal{G}}^\bullet = [\mathbf{D}_p \bar{\mathbf{E}}^\top \bar{S} \quad \mathbf{D}_a \bar{\mathbf{E}}_o^\top \bar{P}]. \quad (9)$$

The incidence matrices

For a directed graph, [1] defines the incidence matrix $\mathbf{E} \in \mathbb{R}^{n \times m}$ and the matrix $\mathbf{E}_o \in \mathbb{R}^{n \times m}$ collecting its outgoing part by

$$\begin{aligned} [\mathbf{E}]_{ik} &= \begin{cases} -1, & \text{if } e_k = (v_i, v_j) \in \mathcal{E} \text{ (outgoing edge)} \\ 1, & \text{if } e_k = (v_j, v_i) \in \mathcal{E} \text{ (ingoing edge)} \\ 0, & \text{otherwise,} \end{cases} \\ [\mathbf{E}_o]_{ik} &= \begin{cases} -1, & \text{if } e_k = (v_i, v_j) \in \mathcal{E} \text{ (outgoing edge)} \\ 0, & \text{otherwise,} \end{cases} \end{aligned} \quad (10)$$

which are eqs. (1) and (2) of [1], together with $\bar{\mathbf{E}} = \mathbf{E} \otimes I_3$ and $\bar{\mathbf{E}}_o = \mathbf{E}_o \otimes I_3$. Transposing turns the k th column into the k th row block, so for an edge $e_k = (v_i, v_j)$, carrying the -1 at its tail v_i and the $+1$ at its head v_j ,

$$[\bar{\mathbf{E}}^\top]_{k,\cdot} = \left[\mathbf{0} \cdots \underbrace{-I_3}_{\text{agent } i} \cdots \underbrace{+I_3}_{\text{agent } j} \cdots \mathbf{0} \right], \quad (11)$$

$$[\bar{\mathbf{E}}_o^\top]_{k,\cdot} = \left[\mathbf{0} \cdots \underbrace{-I_3}_{\text{agent } i} \cdots \underbrace{\mathbf{0}}_{\text{agent } j} \cdots \mathbf{0} \right]. \quad (12)$$

A bearing depends on the positions of *both* endpoints, so $\bar{\mathbf{E}}^\top$ touches both column blocks, whereas it depends on the attitude of the sensing agent only, so $\bar{\mathbf{E}}_o^\top$ touches one.

The edge-indexed product

$\mathbf{D}_p$ and $\mathbf{U}$ are block diagonal over *edges*, so multiplying (11) from the left by $\mathbf{D}_p^{(k)} \mathbf{U}_{ij}$ scales the whole row block,

$$[\mathbf{D}_p \mathbf{U} \bar{\mathbf{E}}^\top]_{k,\cdot} = \left[\mathbf{0} \cdots \underbrace{-\mathbf{D}_p^{(k)} \mathbf{U}_{ij}}_{\text{agent } i} \cdots \underbrace{+\mathbf{D}_p^{(k)} \mathbf{U}_{ij}}_{\text{agent } j} \cdots \mathbf{0} \right], \quad (13)$$

and likewise, using (12),

$$[\mathbf{D}_a \mathbf{V} \bar{\mathbf{E}}_o^\top]_{k,\cdot} = \left[\mathbf{0} \cdots \underbrace{-\mathbf{D}_a^{(k)} \mathbf{V}_{ij}}_{\text{agent } i} \cdots \underbrace{\mathbf{0}}_{\text{agent } j} \cdots \mathbf{0} \right]. \quad (14)$$

Acting on δ^+ this returns

$$(\mathbf{B}^\times \delta^+)_k = \mathbf{D}_p^{(k)} \mathbf{U}_{ij} \delta_{p,j} - \mathbf{D}_p^{(k)} \mathbf{U}_{ij} \delta_{p,i} - \mathbf{D}_a^{(k)} \mathbf{V}_{ij} \delta_{a,i} = \mathbf{D}_p^{(k)} \mathbf{U}_{ij} (\delta_{p,j} - \delta_{p,i}) - \mathbf{D}_a^{(k)} \mathbf{V}_{ij} \delta_{a,i}, \quad (15)$$

which agrees with (5) when $\mathbf{U}_{ij}$ acts as the identity on the relative motion. Comparing the last two expressions in (15), the same matrix $\mathbf{D}_p^{(k)} \mathbf{U}_{ij}$ multiplies $\delta_{p,j}$ and $\delta_{p,i}$, and it may equivalently be read as acting on their difference. That is the whole of Lemma 1 below.

Note also from (14) that $\mathbf{V}_{ij}$ meets only agent i . Setting $\mathbf{V}_{ij} = P_i$ restricts only the agent whose columns it occupies and does not cause problems. Everything below concerns $\mathbf{U}$ alone.

The node-indexed product

Now multiply (11) from the *right* by $\bar{S} = \text{blkdiag}(S_1, \dots, S_n)$, which is block diagonal over *nodes*, so each column block is scaled by its own projector,

$$[\bar{\mathbf{E}}^\top \bar{S}]_{k,\cdot} = \left[\mathbf{0} \cdots -S_i \cdots +S_j \cdots \mathbf{0} \right], \quad [\mathbf{D}_p \bar{\mathbf{E}}^\top \bar{S}]_{k,\cdot} = \left[\mathbf{0} \cdots -\mathbf{D}_p^{(k)} S_i \cdots +\mathbf{D}_p^{(k)} S_j \cdots \mathbf{0} \right], \quad (16)$$

and in the same way $[\mathbf{D}_a \bar{\mathbf{E}}_o^\top \bar{P}]_{k,\cdot}$ has the single nonzero block $-\mathbf{D}_a^{(k)} P_i$ in the columns of agent i . Hence

$$(\mathbf{B}^\bullet \delta^+)_k = \mathbf{D}_p^{(k)} (S_j \delta_{p,j} - S_i \delta_{p,i}) - \mathbf{D}_a^{(k)} P_i \delta_{a,i}. \quad (17)$$

Comparing (15) with (17), the difference is only whether the projector is applied before or after the two endpoint variations are subtracted. If $S_i = S_j = S$, then $S_j \delta_{p,j} - S_i \delta_{p,i} = S(\delta_{p,j} - \delta_{p,i})$ and the two row blocks agree. This condition is weaker than homogeneity, since only the translational projector enters it. An edge joining an $\mathbb{R}^3 \times S^1$ agent to an SE(3) one has $S_i = S_j = I_3$, and an edge joining an $\mathbb{R}^2$ agent to an $\mathbb{R}^2 \times S^1$ one has $S_i = S_j = \text{diag}(1, 1, 0)$, so both are covered. Proposition 2 states the general case.

The two requirements

Lemma 1 (the two endpoints share one factor). *By (13), the column blocks of agents i and j inside row block k of (8) are $-\mathbf{D}_p^{(k)} \mathbf{U}_{ij}$ and $+\mathbf{D}_p^{(k)} \mathbf{U}_{ij}$. Any restriction encoded by $\mathbf{U}_{ij}$ is therefore imposed on both endpoints at once.*

Two requirements are placed on the column blocks of the bearing rigidity matrix. The matrix should reproduce (5) on motions the agents can perform, and the text preceding Theorem 2 in [1] asks that inadmissible degrees of freedom index null columns of the matrix (so that the count q_v is a property of the manifolds alone and cancels between $\text{rk } \mathbf{B}_G^\times = 6n - q_v - q_i$ and $\text{rk } \mathbf{B}_K^\times = 6n - q_v - q_t$). Written for a candidate matrix $\mathbf{B} \in \mathbb{R}^{3m \times 6n}$, the two requirements read

$$\mathbf{B}A = J_G, \quad (\text{F})$$

$$\mathbf{B}(I - AA^\top) = 0. \quad (\text{Z})$$

Here q_i and q_t denote the dimensions of the infinitesimal and the trivial variation sets of [1], both read in the coordinates of δ , so that $q_i = \dim \ker(\mathbf{B}_G A)$ and $q_t = \dim \ker(\mathbf{B}_K A)$.

Definition 13 of [1] determines a matrix only on $\mathcal{A}$, since δ^+ never leaves that subspace, so it does not single out either construction. Both $\mathbf{B}_G^\times$ and $\mathbf{B}_K^\times$ satisfy it, and they differ only on $\mathcal{A}^\perp$. Condition (F) is Definition 13 restated, and (Z) is what decides between them.

Lemma 2 (when (F) and (Z) are compatible). *Let $e_k = (v_i, v_j)$ with i planar ($S_i = \text{diag}(1, 1, 0)$) and j spatial ($S_j = I_3$). Then $\mathbf{B}_G^\times$ satisfies (F) and (Z) on the column blocks of i and j only if $P(\hat{p}_{ij})e_3 = 0$, that is, only if agent j lies vertically above or below agent i .*

Proof. Write $U = \mathbf{U}_{ij}$ and read the two column blocks of (13) in turn.

Agent j . Its admissible directions are all of $\mathbb{R}^3$, and by (17) the block that reproduces (5) there is $+\mathbf{D}_p^{(k)}S_j = \mathbf{D}_p^{(k)}$. So (F) reads

$$\mathbf{D}_p^{(k)}Ux = \mathbf{D}_p^{(k)}x \quad \text{for all } x \in \mathbb{R}^3, \quad \text{i.e.} \quad \mathbf{D}_p^{(k)}U = \mathbf{D}_p^{(k)}. \quad (18)$$

Agent i . Its inadmissible direction is e_3 , and by (13) the corresponding column of the matrix is $-\mathbf{D}_p^{(k)}Ue_3$. So (Z) reads

$$\mathbf{D}_p^{(k)}Ue_3 = 0. \quad (19)$$

Substituting (18) into (19) gives $\mathbf{D}_p^{(k)}e_3 = 0$, that is $e_3 \in \ker \mathbf{D}_p^{(k)}$. Finally

$$\mathbf{D}_p^{(k)} = s_{ij}R_i^\top P(\hat{p}_{ij}), \quad s_{ij} \neq 0, \quad R_i \text{ orthogonal}, \quad \text{so} \quad \ker \mathbf{D}_p^{(k)} = \ker P(\hat{p}_{ij}) = \text{span}\{\hat{p}_{ij}\},$$

and $e_3 \in \text{span}\{\hat{p}_{ij}\}$ says $e_3 = c\hat{p}_{ij}$ for some $c \in \mathbb{R}$. Both vectors have unit norm, so $c = \pm 1$ and $\hat{p}_{ij} = \pm e_3$. $\square$

Running the same two steps with general S_i, S_j gives $\mathbf{D}_p^{(k)}U = \mathbf{D}_p^{(k)}S_i$ from agent i and $\mathbf{D}_p^{(k)}U = \mathbf{D}_p^{(k)}S_j$ from agent j , hence

$$\mathbf{D}_p^{(k)}(S_i - S_j) = 0, \quad \text{i.e.} \quad \text{range}(S_i - S_j) \subseteq \text{span}\{\hat{p}_{ij}\}, \quad (20)$$

a nontrivial algebraic condition on (p_i, p_j) whenever $S_i \neq S_j$.

Remark 1 (why this stays invisible until two graphs are compared). By (F) both constructions have the same rank once restricted to admissible columns, $\text{rk}(\mathbf{B}^\times A) = \text{rk}(\mathbf{B}^\bullet A) = \text{rk } J$. They differ only in how much rank the inadmissible columns add, and that amount depends on the edge set. Theorem 2 compares $\text{rk } \mathbf{B}_G$ with $\text{rk } \mathbf{B}_K$, so a difference in that amount is read as a difference in rigidity.

4 Analysis of the example network

The formation (1) has $c = 7$ degrees of freedom, two horizontal components for each of the planar agents and three for agent 3. Write a variation as $\delta = (u_1, u_2, u_3)$ with $(u_1)_3 = (u_2)_3 = 0$ and $u_3 \in \mathbb{R}^3$.

No agent rotates and $R_i = I_3$, so (5) reduces to $\dot{b}_{ij} = s_{ij}P(\hat{p}_{ij})(u_j - u_i)$, and $\ker P(x) = \text{span}\{x\}$. A bearing is preserved to first order exactly when its two endpoints separate along the line joining them. For the two edges of $\mathcal{G}$,

$$u_3 - u_1 = \alpha(p_3 - p_1) = \alpha(1, 1, 1), \quad u_3 - u_2 = \beta(p_3 - p_2) = \beta(-1, 1, 1), \quad \alpha, \beta \in \mathbb{R}, \quad (21)$$

where α and β are rates, the spanning vector being taken unnormalised so that the two third components are both 1.

Third components. Both $p_3 - p_1$ and $p_3 - p_2$ have third component 1, while u_1 and u_2 have none. Reading the third row of each equation in (21) therefore gives $\alpha = (u_3)_3$ and $\beta = (u_3)_3$, so the two scalars are equal. Call the common value t .

Both equations at once. With $\alpha = \beta = t$, (21) reads $u_i = u_3 - t(p_3 - p_i)$ for $i = 1, 2$, that is

$$u_i = w + t p_i \quad (i = 1, 2, 3), \quad w := u_3 - t p_3, \quad (22)$$

which describes an infinitesimal similarity, translating the formation by w and scaling it by t about the origin.

Which similarities are admissible. Since $p_1 = 0$ we get $w = u_1$, which is horizontal. Conversely any (w, t) with w horizontal is admissible, because $u_2 = w + t p_2$ is horizontal, p_2 being horizontal, and u_3 is unconstrained. The kernel is thus 3-dimensional, spanned by the two horizontal translations and the uniform scaling.

The same for $\mathcal{K}$. A similarity (22) preserves *every* bearing of the formation, since $u_j - u_i = t(p_j - p_i)$ for all i, j , not only for the two edges of $\mathcal{G}$. Hence $\ker J_{\mathcal{G}} = \ker J_{\mathcal{K}}$, the framework is IBR in the sense of Definition 12 of [1], and

$$\text{rk } J_{\mathcal{G}} = \text{rk } J_{\mathcal{K}} = c - 3 = 4.$$

Two edges, each contributing rank 2, attain this, so $\mathcal{G}$ is minimally rigid.

For $e_1 = (v_1, v_3)$ we have $p_3 - p_1 = (1, 1, 1)$ of norm $\sqrt{3}$ and $s_{13} = 1/\sqrt{3}$, so

$$\mathbf{D}_p^{(1)} = s_{13}(I_3 - \hat{p}_{13}\hat{p}_{13}^\top) = \frac{1}{\sqrt{3}} \left(I_3 - \frac{1}{3} \begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix} \right) = \frac{\sqrt{3}}{9} \begin{bmatrix} 2 & -1 & -1 \\ -1 & 2 & -1 \\ -1 & -1 & 2 \end{bmatrix},$$

and for $e_2 = (v_2, v_3)$ we have $p_3 - p_2 = (-1, 1, 1)$, again of norm $\sqrt{3}$, so

$$\mathbf{D}_p^{(2)} = \frac{\sqrt{3}}{9} \begin{bmatrix} 2 & 1 & 1 \\ 1 & 2 & -1 \\ 1 & -1 & 2 \end{bmatrix}.$$

Placing these by (13) with $\mathbf{U}_{ij} = I_3$, and dropping the attitude half which is identically zero here,

$$\mathbf{B}_{\mathcal{G}}^\times = \frac{\sqrt{3}}{9} \left[\begin{array}{ccc|ccc|ccc} -2 & 1 & 1 & 0 & 0 & 0 & 2 & -1 & -1 \\ 1 & -2 & 1 & 0 & 0 & 0 & -1 & 2 & -1 \\ 1 & 1 & -2 & 0 & 0 & 0 & -1 & -1 & 2 \\ \hline 0 & 0 & 0 & -2 & -1 & -1 & 2 & 1 & 1 \\ 0 & 0 & 0 & -1 & -2 & 1 & 1 & 2 & -1 \\ 0 & 0 & 0 & -1 & 1 & -2 & 1 & -1 & 2 \end{array} \right], \quad (23)$$

the columns being $(u_{1x}, u_{1y}, u_{1z} \mid u_{2x}, u_{2y}, u_{2z} \mid u_{3x}, u_{3y}, u_{3z})$. The third and sixth columns hold the vertical motions of the two planar agents, and neither is zero. The node-indexed matrix (16) is (23) with those two columns replaced by zeros, since $S_1 = S_2 = \text{diag}(1, 1, 0)$ annihilates them while $S_3 = I_3$ leaves the rest alone.

On $\mathcal{G}$ both constructions have rank 4, as two row blocks of rank 2 cannot give more. The complete graph adds a row block for $e_3 = (v_1, v_2)$, with $\hat{p}_{12} = e_1$ and $s_{12} = \frac{1}{2}$.¹

Here the two constructions part. On the nine lifted columns the kernel of $\mathbf{B}_{\mathcal{K}}^\times$ consists of the four infinitesimal similarities of $\mathbb{R}^3$, three translations and the scaling, so $\text{rk } \mathbf{B}_{\mathcal{K}}^\times = 9 - 4 = 5$. Only three of the four are admissible, since the vertical translation lifts the planar agents out of their plane. The node-indexed matrix has zeros in columns three and six, so its rank is that of the remaining seven columns, $\text{rk } \mathbf{B}_{\mathcal{K}}^\bullet = 7 - 3 = 4$. The comparison of $\mathbf{B}_{\mathcal{G}}^\times$ with $\mathbf{B}_{\mathcal{K}}^\times$ is then 4 against 5 and reports flexibility, where that of $\mathbf{B}_{\mathcal{G}}^\bullet$ with $\mathbf{B}_{\mathcal{K}}^\bullet$ is 4 against 4.

One might suspect that the value 5 comes from having used $\mathbf{U}_{12} = I_3$ on the edge between the two planar agents, where Table III of [1] prescribes $[e_1 \ e_2 \ 0_3]$ for a pair of terrestrial agents. It does not. The z columns of agents 1 and 2 receive entries from three row blocks, one for each edge of $\mathcal{K}$, and $\mathbf{U}_{12}$ governs only one of them. Taking $\mathbf{U}_{12} = [e_1 \ e_2 \ 0_3]$ turns the factor

¹Since all three agents here are particles with $R_i = I_3$, reversing an edge negates its row block and changes no rank or kernel, so $\mathcal{K}$ may be read as oriented with three row blocks or as directed with six. Section V-A1 of [1] takes the graph in the $\mathbb{R}^d$ case to be undirected for the same reason.

$\mathbf{D}_p^{(3)}\mathbf{U}_{12}$ of that row block from $\frac{1}{2}\text{diag}(0, 1, 1)$ into $\frac{1}{2}\text{diag}(0, 1, 0)$, so it contributes nothing to the two columns. The row blocks of the two mixed edges still do, since Table III prescribes $\mathbf{U}_{ij} = I_3$ there, and $\text{rk } \mathbf{B}_\mathcal{K}^\times$ is again 5.

Remark 2 (a second mechanism). In the example, the number of null columns happens to be the same for $\mathcal{G}$ and $\mathcal{K}$, and the discrepancy is entirely in how much rank the nonzero inadmissible columns contribute. The count itself can also become graph-dependent. For $n = 4$ with $\mathcal{D}_{1,2,3} = \mathbb{R}^2$, $\mathcal{D}_4 = \mathbb{R}^3$ at $p_1 = (0, 0, 0)$, $p_2 = (1, 0, 0)$, $p_3 = (1, 1, 0)$, $p_4 = (\frac{1}{2}, \frac{1}{2}, 1)$ and $\mathcal{E} = \{(1, 2), (1, 3), (2, 4), (3, 4)\}$, every edge of $\mathcal{G}$ incident to agent 1 joins it to another planar agent, so by Proposition 2 those row blocks agree with the node-indexed ones, where $S_1 = \text{diag}(1, 1, 0)$ sets the z column of agent 1 to zero. In $\mathcal{K}$ that agent is joined to the spatial one as well, its z column is no longer zero, and there are 13 null columns against 12. That framework is again minimally rigid ($c - q_t = 9 - 3 = 6$, attained by four edges) and again reported flexible, $\text{rk } \mathbf{B}_\mathcal{G}^\times = 6$ against $\text{rk } \mathbf{B}_\mathcal{K}^\times = 8$. Deleting the three planar z columns of $\mathbf{B}_\mathcal{K}^\times$ returns its rank to 6.

Remark 3 (the same arithmetic in Section VI-B of [1]). The heterogeneous case study there reports $q_t = 5$, while the accompanying list of trivial motions has four entries. Two planar translations, one coordinated rotation, one scaling. The fifth is the global z translation, which is not admissible for the planar unicycles. Using the entries of Table III of [1], and generic positions since the article does not give the configuration, we reproduce the reported $\text{rk } \mathbf{B}_\mathcal{K}^\times = 13$ with six null columns. Deleting the three unicycle z columns lowers it to $11 = c - q_t = 15 - 4$. The graph there is complete, so the rigidity conclusion of the case study is unaffected either way.

5 Analysis of the node-indexed construction

Proposition 1 (uniqueness). *There is exactly one matrix satisfying (F) and (Z), namely $J_\mathcal{G}A^\top$, and it equals $\mathbf{B}_\mathcal{G}^\bullet$ of (9).*

Proof. Since $\bar{S}$ and $\bar{P}$ are orthogonal projectors and $\mathcal{A}$ splits along the position and the attitude coordinate blocks, the orthogonal projector onto $\mathcal{A}$ is

$$AA^\top = \text{blkdiag}(\bar{S}, \bar{P}). \quad (24)$$

Write $\mathbf{M} := \begin{bmatrix} \mathbf{D}_p \bar{\mathbf{E}}^\top & \mathbf{D}_a \bar{\mathbf{E}}_o^\top \end{bmatrix}$ for the differential before any restriction, so that (9) reads $\mathbf{B}_\mathcal{G}^\bullet = \mathbf{M} \text{blkdiag}(\bar{S}, \bar{P}) = \mathbf{M}AA^\top$ by (24). Idempotency of $AA^\top$ and $A^\top A = I_c$ then give

$$\mathbf{B}_\mathcal{G}^\bullet(I - AA^\top) = \mathbf{M}AA^\top(I - AA^\top) = 0, \quad \mathbf{B}_\mathcal{G}^\bullet A = \mathbf{M}AA^\top A = \mathbf{M}A = J_\mathcal{G},$$

the last equality because $\mathbf{M}$ reproduces (5) on admissible variations. So $\mathbf{B}_\mathcal{G}^\bullet$ satisfies (F) and (Z). Conversely, let $\mathbf{B}$ satisfy both. From $I_{6n} = AA^\top + (I_{6n} - AA^\top)$ we obtain

$$\mathbf{B} = \mathbf{B}AA^\top + \mathbf{B}(I_{6n} - AA^\top) = (\mathbf{B}A)A^\top = J_\mathcal{G}A^\top,$$

so the matrix is unique and equals $\mathbf{B}_\mathcal{G}^\bullet$. $\square$

The construction $\mathbf{B}_\mathcal{G}^\bullet$ completes Definition 13 of [1] to the virtual subspace that the null-column requirement of Theorem 2 asks for, and by Proposition 1 it is the only one.

Theorem 1. *For every $\mathcal{G}$, $\text{rk } \mathbf{B}_\mathcal{G}^\bullet = 6n - q_v - q_i$ with $q_v = 6n - c$ and $q_i = \dim \ker(\mathbf{B}_\mathcal{G}^\bullet A)$. Hence $\text{rk } \mathbf{B}_\mathcal{G}^\bullet = \text{rk } \mathbf{B}_\mathcal{K}^\bullet$ if and only if $q_i = q_t$, that is, if and only if $(\mathcal{G}, \chi)$ is IBR.*

Proof. By (Z), $\mathbf{B}_{\mathcal{G}}^\bullet = \mathbf{B}_{\mathcal{G}}^\bullet A A^\top$, so the row space is unchanged by restricting to the admissible columns and

$$\text{rk } \mathbf{B}_{\mathcal{G}}^\bullet = \text{rk}(\mathbf{B}_{\mathcal{G}}^\bullet A) = \underbrace{c}_{\text{columns of } A} - \underbrace{q_i}_{\dim \ker} = (6n - q_v) - q_i.$$

The same holds for $\mathcal{K}$ with $q_t = \dim \ker(\mathbf{B}_{\mathcal{K}}^\bullet A)$ and with the *same* q_v , since by (6) it depends only on the manifolds. Subtracting, $\text{rk } \mathbf{B}_{\mathcal{G}}^\bullet - \text{rk } \mathbf{B}_{\mathcal{K}}^\bullet = q_t - q_i$, so the two ranks agree if and only if $q_i = q_t$. By Theorem 1 of [1], $\ker \mathbf{B}_{\mathcal{K}} \subseteq \ker \mathbf{B}_{\mathcal{G}}$, so equality of the two dimensions is equality of the two kernels, which is Definition 12 of [1]. $\square$

Proposition 2 (when the two constructions coincide). *Take $\mathbf{U}_{ij} = S_i$ and $\mathbf{V}_{ij} = P_i$. For an edge $e_k = (v_i, v_j)$ with $S_i = S_j$, the k th row blocks of $\mathbf{B}_{\mathcal{G}}^\times$ and $\mathbf{B}_{\mathcal{G}}^\bullet$ agree. Hence if $S_i = S_j$ on every edge of $\mathcal{E}$, then $\mathbf{B}_{\mathcal{G}}^\times = \mathbf{B}_{\mathcal{G}}^\bullet$ at every configuration. Homogeneous formations are the special case $\mathcal{D}_i = \mathcal{D}$ for all i , where $\mathbf{U}_{ij} = S$ and $\mathbf{V}_{ij} = P$ are the entries of Table I of [1].*

Proof. Compare (15) and (17) on row block k . Writing S for the common value of S_i and S_j ,

$$\mathbf{D}_p^{(k)} S (\delta_{p,j} - \delta_{p,i}) = \mathbf{D}_p^{(k)} (S \delta_{p,j} - S \delta_{p,i}) = \mathbf{D}_p^{(k)} (S_j \delta_{p,j} - S_i \delta_{p,i})$$

by linearity of S , and the attitude blocks agree already, since $\bar{\mathbf{E}}_o^\top$ meets only agent i and $\mathbf{V}_{ij} = P_i$. Applying this to every k gives the second claim. In the homogeneous case $\bar{S} = I_n \otimes S$ commutes past the incidence matrix, $\mathbf{U} \bar{\mathbf{E}}^\top = \bar{\mathbf{E}}^\top \bar{S}$ and $\mathbf{V} \bar{\mathbf{E}}_o^\top = \bar{\mathbf{E}}_o^\top \bar{P}$.² $\square$

Remark 4 (what the condition asks of $\mathcal{G}$ and of $\mathcal{K}$). The hypothesis of Proposition 2 is about the edges present in $\mathcal{G}$, which is where the graph dependence enters. Since $\mathcal{K}$ contains every pair, $\mathbf{B}_{\mathcal{K}}^\times = \mathbf{B}_{\mathcal{K}}^\bullet$ requires $S_i = S_j$ for all i and j , so any formation mixing planar with spatial agents fails it on the complete graph while $\mathcal{G}$ itself may satisfy it. The converse implication does not hold at the level of ranks. In Section 4 both edges of $\mathcal{G}$ join a planar to a spatial agent, so $\mathbf{B}_{\mathcal{G}}^\times \neq \mathbf{B}_{\mathcal{G}}^\bullet$, and the two still have the same rank 4.

Implementation. By (16) the change is one line per block. For row block $k = (i, j)$, each index naming the top-left corner of a 3×3 block,

$$\mathbf{B}^\bullet[3k, 3j] += \mathbf{D}_p^{(k)} S_j, \quad \mathbf{B}^\bullet[3k, 3i] -= \mathbf{D}_p^{(k)} S_i, \quad \mathbf{B}^\bullet[3k, 3n+3i] += R_i^\top [\hat{p}_{ij}] \times P_i,$$

the last line being $-\mathbf{D}_a^{(k)} P_i$ written out via (5). The two dense $3m \times 3m$ factors are no longer needed. One may also avoid the lift altogether and apply the rank test to the c -column matrices $J_{\mathcal{G}} = \mathbf{B}_{\mathcal{G}}^\bullet A$ and $J_{\mathcal{K}}$, in which case q_v never enters the bookkeeping at all.

Relation to the heterogeneous formulation of Pozzan et al. The node-wise treatment is not new. It is used in [2], where each agent carries its own selection matrices $S_{p,i} \in \mathbb{R}^{3 \times c_{p,i}}$ and $S_{o,i} \in \mathbb{R}^{3 \times c_{o,i}}$ mapping its command vector to its linear and angular velocity, assembled block-diagonally into $\mathbf{S}_p = \text{diag}(S_{p,i})$ and $\mathbf{S}_o = \text{diag}(S_{o,i})$ in eqs. (3), (4) and (7) there. The bearing rigidity matrix defined there is $\mathbf{B}_{\mathcal{G}}(\mathbf{x}) \in \mathbb{R}^{3m \times c}$, and its k th row block, eq. (9) of [2], carries $S_{p,i}$ in the column block of agent i and $S_{p,j}$ in the column block of agent j . The two endpoints are restricted separately, which is what (16) does here. Infinitesimal rigidity is then defined by $\ker \mathbf{B}_{\mathcal{K}} = \ker \mathbf{B}_{\mathcal{G}}$ on those c columns, so neither the lift nor the count q_v ever appears.

The correspondence with (9) is $J_{\mathcal{G}} = \mathbf{B}_{\mathcal{G}}^\bullet A$, with $\text{blkdiag}(\mathbf{S}_p, \mathbf{S}_o)$ in the role of A . Note that $S_{p,i}$ is an injection whose columns span the admissible directions rather than a projector, so it plays the role of A and not of $\bar{S}$, the relation being $S_i = S_{p,i} S_{p,i}^\top$ for orthonormal columns.

²For $\mathbb{R}^3 \times S^1$ read $\mathbf{V}_{ij}$ in the projector form $P_i = vv^\top$ rather than the $[0_{3 \times 2} \ v]$ form of Table I of [1]. The two differ as matrices but not in rank, nor in q_v . Only the phrase “null column” is sensitive to the choice, which is why (Z) is stated as an identity.

The two formulations differ also in that [2] parametrises the commands in the body frame, $\dot{p}_i = R_i v_i$, which is what puts $R_i^\top R_j S_{p,j}$ in the column block of agent j , whereas S_i and P_i here are world-frame projectors. What the present note adds is that this choice is forced rather than convenient. By Lemma 2 the same restriction cannot be expressed in $\mathbf{B}_G^\times$, and by Proposition 1 $\mathbf{B}_G^\bullet$ is the only matrix compatible with the null-column requirement that Theorem 2 of [1] uses.

6 Numerical checks

The reference below is the Jacobian of the bearing map along admissible variations, obtained by central differences on the manifold with attitudes updated as $R_i \mapsto \exp([\varepsilon \omega_i]_\times) R_i$. It uses neither construction.

Over 432 frameworks covering the five manifolds at $n = 3, 4, 6, 9$ and sixteen heterogeneous mixes at four graph densities, both (8) and (9) reproduce this Jacobian on $\mathcal{A}$ to a worst relative error of 1.1×10^{-9} , as (F) requires of both. The identity $\mathbf{B}^\bullet(I - AA^\top) = 0$ holds to 2.2×10^{-16} , and the corresponding quantity for (8) is of order one. Over 1200 random heterogeneous frameworks with $n \in \{4, 5, 6\}$ we found the following.

	edge-indexed	node-indexed
$\text{rk } \mathbf{B} = 6n - q_v - q_i$ fails	504/1198	0/1198
rank test disagrees with the Jacobian	67/1198 (5.6%)	0/1198

For homogeneous formations the two constructions agree bitwise, checked over 25 graphs in each manifold at $n = 3, 4, 6, 8, 12$, and the ranks match the closed forms $2n - 3$, $3n - 4$, $3n - 4$, $4n - 5$, $6n - 7$. Reproducing the case study of Section VI-B of [1] with (8) returns $\text{rk } \mathbf{B}_K^\times = 13$ with six null columns, the values reported there, which suggests that the difference lies in the construction rather than in our implementation of it.

References

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